

Revised Generation Interconnection System Impact Study Report

For Queue Project Q20-1606

Olympus 138 kV
October 27, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1606. The Interconnection Customer proposes a 202.0 MW solar photovoltaic facility interconnecting at the Olympus 138 kV substation in the Ohio area. The request was received on November 11, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 599 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 8 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$41,038,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Leon to Eagle Mountain 138.0 kV	Olympus 230 kV/1 kV XFMR	133.8	223	54.2
Villas De Castro 3013 to Ravenswood 138.0 kV	Sporn to Gavin Power Llc 230.0 kV	116.7	200	12.9
Cottageville to Villas De Castro 3013 138.0 kV	Gavin Power Llc to Sporn 230.0 kV	113.3	208	12.9
Cottageville to North Attleboo 138.0 kV	Gavin Power Llc to Sporn 230.0 kV L	113.3	211	12.9
Eaymond to N Attleboro 138.0 kV	Sporn to Gavin Power Llc 230.0 kV	111.9	218	12.9
Leon Substation to Eaymond 138.0 kV	Sporn to Gavin Power Llc 230.0 kV Line	111.0	224	12.9
Ossining to Pomona Park 138.0 kV	Olympus 230/1 kV XFMR	106.8	238	80.1
Ossining to Eagle Mtn 138.0 kV	Olympus 230/1kV Auto	105.5	232	80.1

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Leon to Eagle Mtn 138.0 kV Line	Rebuild line with higher-capacity conductor	\$9,697,000	79% (\$7,677,000)
Villas De Castro 3013 to Ravenswood 138.0 kV	Rebuild line with higher-capacity conductor	\$8,809,000	100% (\$8,809,000)
Cottageville to Villas De Castro 3013 138.0 kV	Rebuild line with higher-capacity conductor	\$3,226,000	100% (\$3,226,000)
Cottageville to North Attleboo 138.0 kV	Rebuild line with higher-capacity conductor	\$8,621,000	100% (\$8,621,000)
N Attleboro to Eaymond 138.0 kV	Rebuild line with higher-capacity conductor	\$7,095,000	100% (\$7,095,000)
Eaymnd to Leon 138.0 kV	Rebuild line with higher-capacity conductor	\$3,521,000	100% (\$3,521,000)
Ossining to Pomona Park 138.0 kV	Reconductor line	\$1,002,000	83% (\$831,000)
Ossining to Eagle Mn 138.0 kV line	Reconductor line	\$1,516,000	83% (\$1,258,000)
	Total allocated to project		\$41,038,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.